VLAB User Guide

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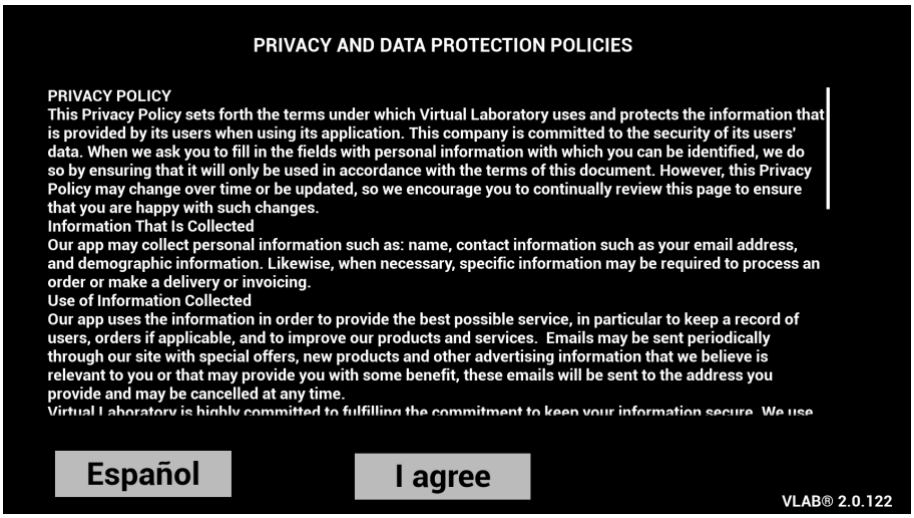
Overview

VLAB is a cross-platform virtual laboratory application designed to deploy temporally precise behavioral experiments. The platform operates on a dual-access model, strictly distinguishing between two user roles:

- **Experimenters:** Users authorized to construct, parameterize, and manage experimental paradigms.
- **Participants:** Users accessing the application strictly to execute tasks and transmit behavioral data.

Getting Started

1. Initial Launch and Privacy Consent Upon launching the VLAB application for the first time, all users will be prompted with the platform's data handling and privacy terms. You must review and accept the Privacy Policy to proceed to the main interface.



2. Authentication Key Entry On the home screen, enter your assigned VLAB access KEY. This unique key is required to authenticate your session and automatically determines your access permissions (Experimenter vs. Participant). If you do not yet have an access key, you must request one from the VLAB administrative team a keyrequest@vlab.website.

3. Role Assignment and Interface Loading Once the key is validated by the server, your access role is automatically assigned. The application will dynamically load either the **Experimenter Dashboard** (for designing and managing tasks) or the **Participant Interface** (for executing a scheduled task), depending on the permissions linked to your specific key.

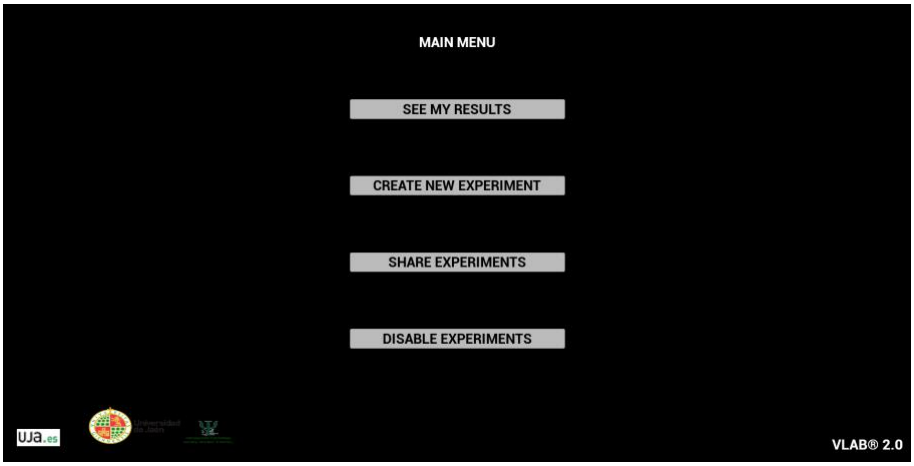
Experimenter Guide

Interface Routing

Initial Launch (First-Time Users): > Upon authenticating for the first time, VLAB will automatically bypass the Main Menu and direct you straight to the **Create Experiment** interface (detailed below).

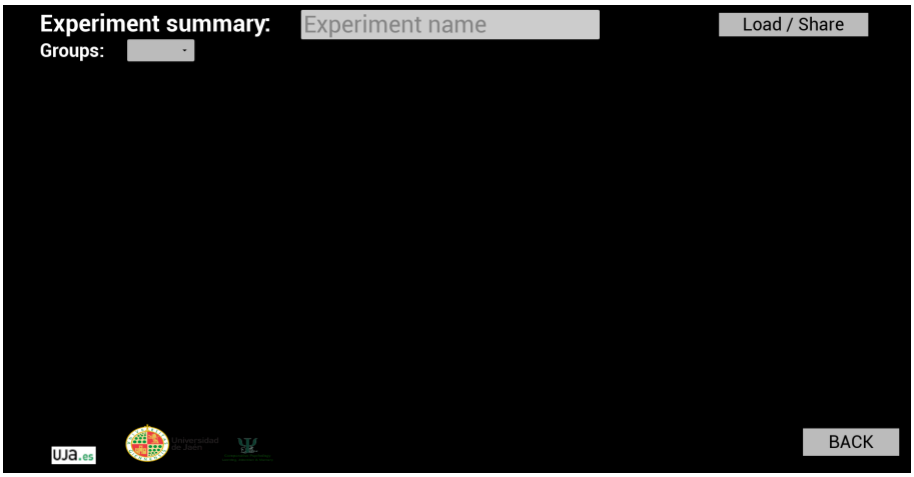
Subsequent Launches: On all future logins, you will be directed to the standard **Main Menu**. From here, you can manage existing projects or start new ones.

Main Menu

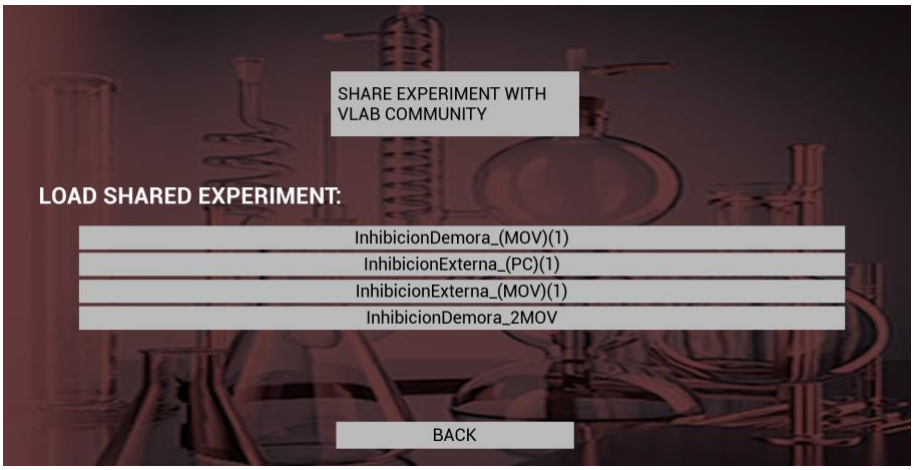


1. Creating an Experiment

1. Enter Experiment Name In the designated text field, input a unique name or identifier for your new experiment. *Note: Ensure the name is distinct from your previous projects to prevent data-logging conflicts on the central server.*

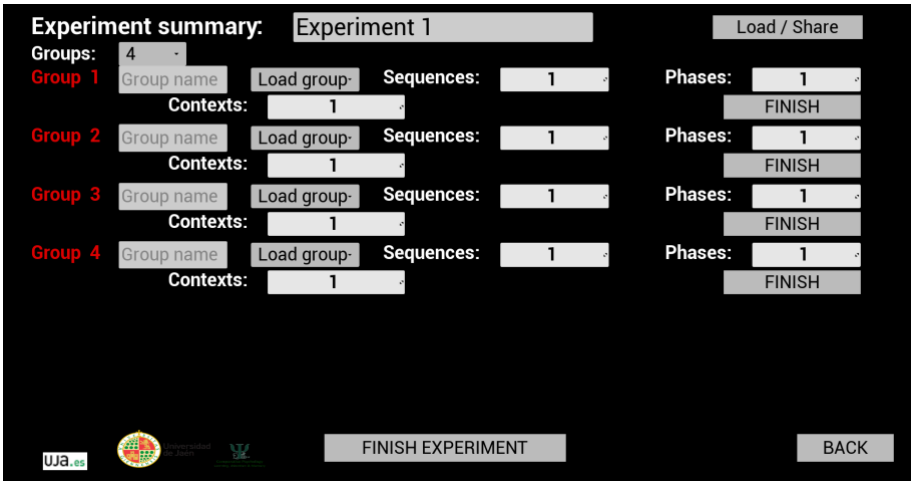


1.1. Loading Existing Configurations Alternatively, instead of starting a new project from scratch, you can access pre-built experimental templates or load the parameters of your previously saved experiments by clicking the **Load Experiment** button.



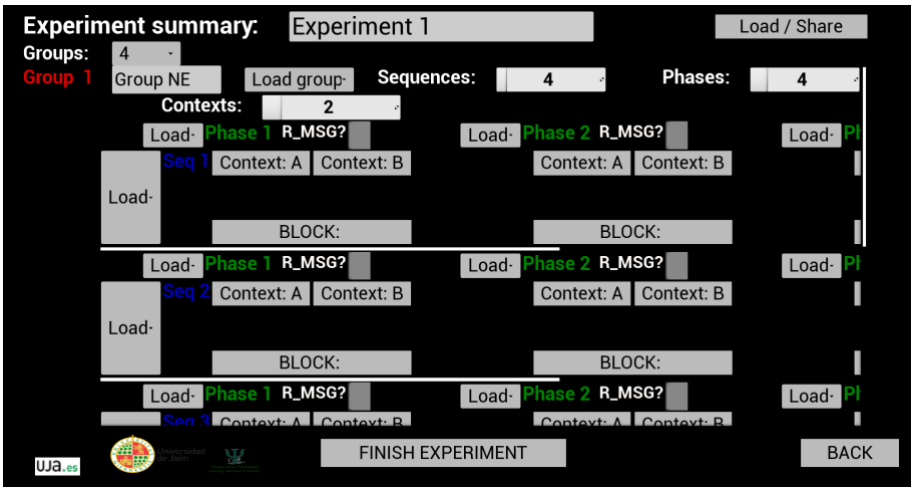
2. Define Experimental Structure Using the provided numerical input fields, select the number of groups, phases, and sequences that will comprise your experimental design. These parameters establish the foundational architecture of your task:

- **Groups:** The number of independent between-subjects conditions (e.g., Control vs. Experimental).
- **Phases:** The distinct temporal stages of the experiment (e.g., Baseline, Acquisition, Reversal, Extinction).
- **Sequences:** The specific ordering, blocks, or pseudo-randomized arrangements of trials presented within those phases.



3. Configure Sequences, Phases, and Contexts Once your experimental structure is established, navigate through the newly generated configuration menus to parameterize the specific details of your task:

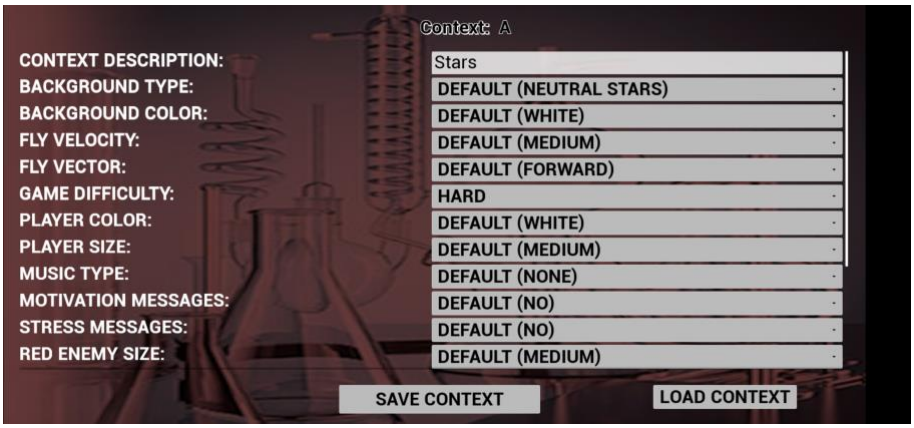
- **Sequences:** Define the specific trial types, stimulus presentation order, and timing intervals within your blocks.
- **Phases:** Establish the rules, transition criteria, and duration for each distinct stage of the experiment.
- **Contexts:** Set the environmental background (e.g., visual scenery, screen color, or ambient auditory cues) present during specific phases or trials.



Configuring Contexts

Within the Contexts menu, you can customize the global environment and gamified elements of the VLAB task. Adjusting these parameters allows you to manipulate the background environment or alter the operant baseline. The following elements can be customized:

- **Background:** Select the visual scenery or color palette that serves as the overarching environmental context for the experimental phase.
- **Player Appearance:** Modify the visual representation of the participant's controllable avatar (e.g., the size).
- **Difficulty:** Adjust baseline challenge metrics, such as the overall speed or enemy attack rate of standard incoming targets.
- **Music:** Select ambient auditory stimuli to serve as contextual background noise or gamified soundtracks.
- **Rewards:** Configure the point values and auditory cues associated with successful baseline operant responses (e.g., destroying standard enemies).



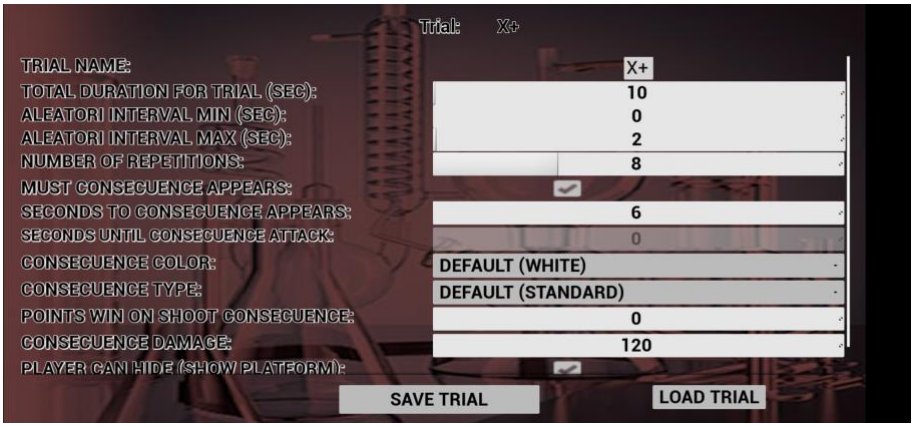
Configuring Trials

Within the Trials menu, you define the core associative learning parameters, timing variables, and specific stimulus presentations for your experimental design. The following elements can be configured:

- **Duration:** Set the exact presentation length (in seconds) of the warning signal.
- **Intervals:** Define the timing rules for the Inter-Trial Interval (ITI), including establishing fixed or variable durations between subsequent trial presentations.
- **Repetitions:** Specify the number of times a distinct trial type is presented within a given sequence or block.
- **Kraken Appearance:** Configure the presentation of the aversive outcome (internally referred to as the "Kraken" or monster mothership). This includes

setting its onset time, duration, attack trajectory (e.g., laser spread), and the corresponding point penalty.

- **Player Abilities:** Define the participant's operational mechanics during the specific trial, such as enabling or disabling their ability to access the green safe zones.



Configuring Signals

Within the Signals menu, you can customize the specific sensory properties of the warning signals—the Conditioned Stimuli (CS)—presented to the participant, as well as the overarching user interface for performance feedback. The following elements can be configured:

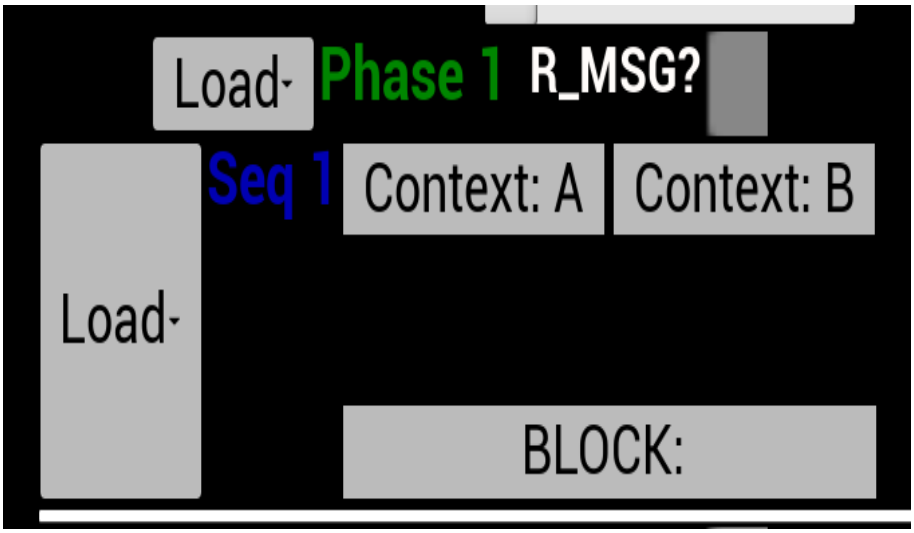
- **Audio:** Select or configure specific sound files to serve as auditory warning signals or to accompany visual cues.
- **Visual:** Define the visual characteristics of the warning signals, such as assigning specific colors (e.g., blue, yellow, pink, or cyan) to the on-screen sensors that alert the participant.
- **Points Position on Screen:** Adjust the spatial location of the participant's total score display within the game interface to ensure it does not obscure critical task stimuli or the safe zones.
- **Points Color:** Customize the font color of the score counter to maintain visual contrast against your selected background context.

Duplicating Configurations (Sequences, Phases, Contexts, or Trials)

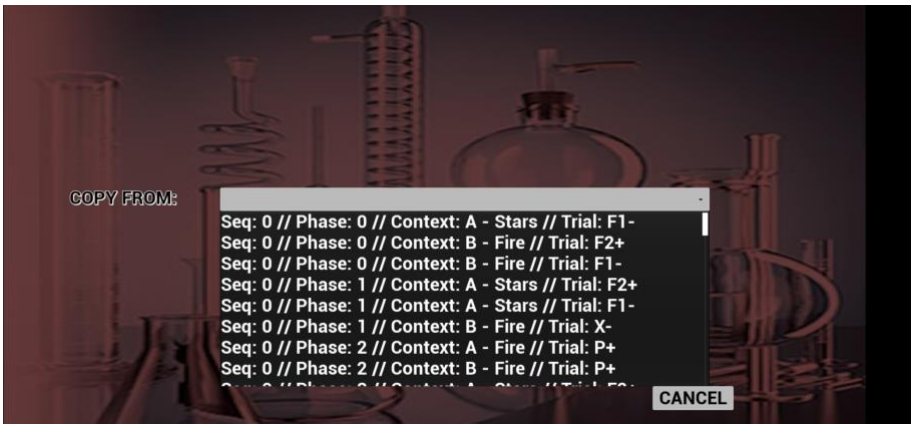
To streamline the design process and maintain consistency across complex experiments, VLAB allows you to duplicate existing parameter sets rather than configuring every element from scratch.

1. Navigate to the Destination: Open the new, blank menu screen for the specific element you are currently building (e.g., a new Phase or Trial).

2. Initiate the Import: Click the **Load** button located within that active menu.

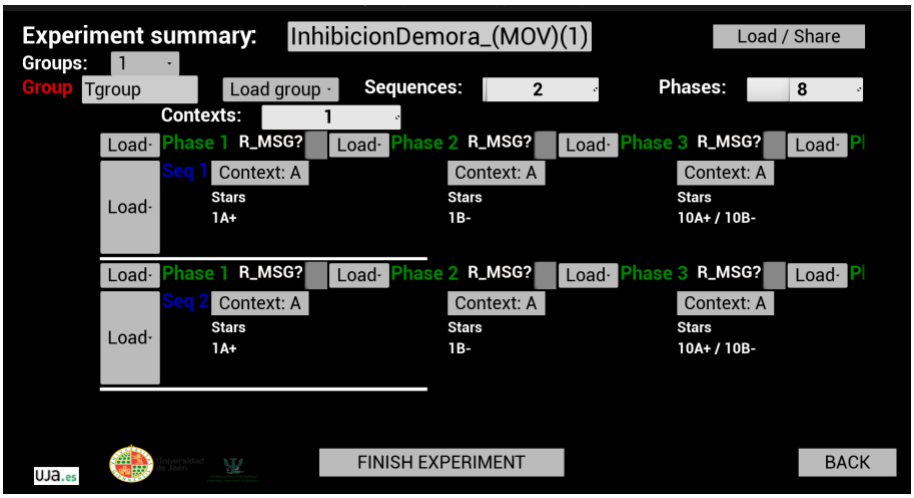


3. Select the Source Configuration: From the prompted menu, select the specific, previously saved file or template that you want to load. Once selected, its parameters will automatically populate your current screen, where they can be further edited or saved as a new version.



Final Setup

Completing the Configuration: Once all required fields and experimental parameters (Groups, Sequences, Phases, Contexts, and Trials) have been fully configured across your design:

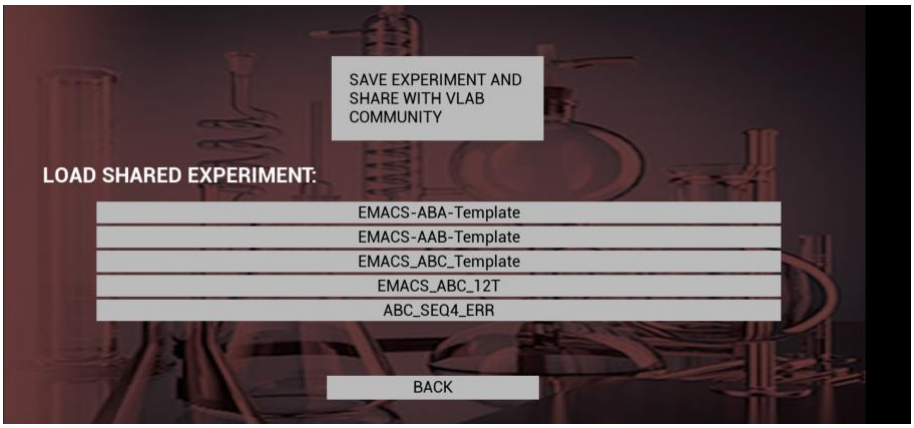


Saving and Publishing Your Experiment

Once you have finished configuring all parameters of your experimental design, you must choose how to store and distribute your task. You have two options:

- **Option 1: Publish to the VLAB Community (Public)** > To share your paradigm with other researchers, navigate to the **Load / Share** Experiment menu, save your current configuration, and select the option to share it with the VLAB community. This adds your experiment to the public repository, making it permanently accessible to all app users (including yourself) for collaborative use.

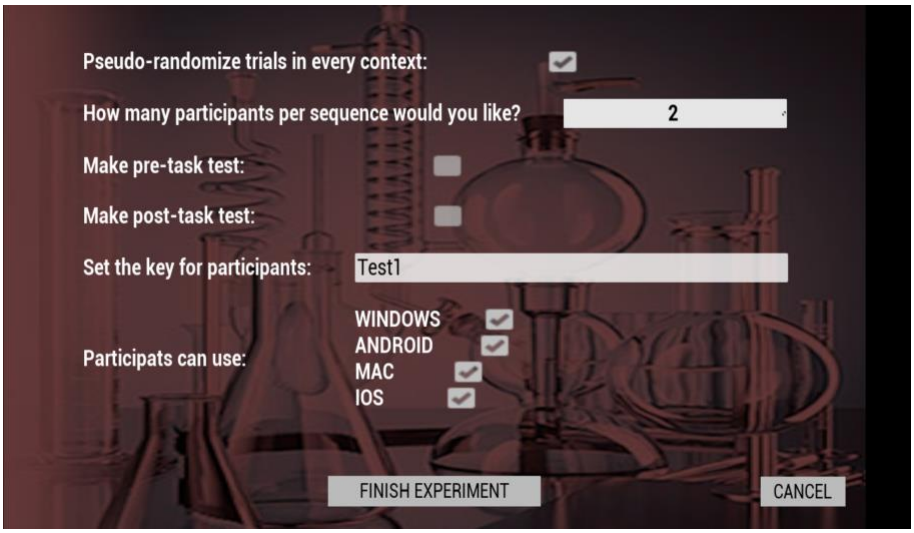
- **Option 2: Save to Personal Account (Private)** If you prefer to keep your research design private, simply click the **Finish Experiment** button. This will bypass the public repository and save the experimental configuration exclusively to your personal researcher account.



Finalizing and Deploying the Experiment

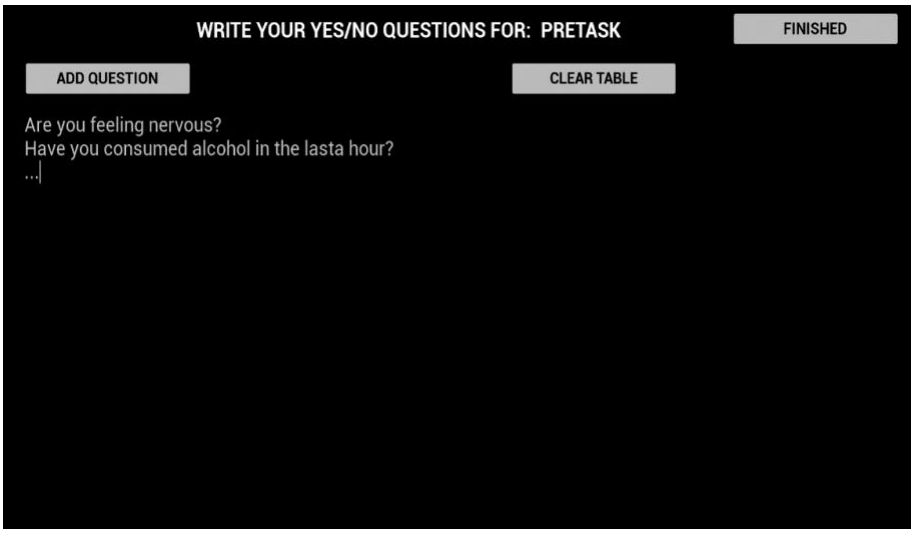
After completing your design and clicking the **Finish Experiment** button, you will be directed to the deployment screen. Here, you must configure the final logistical parameters before distributing the task to your participants:

- **Randomization:** If enabled, trials will be randomized within each context instead of being presented in the configured order.
- **Participants per Sequence:** Set the target sample size or capacity limit for each sequence to ensure balanced data collection across your experimental conditions.
- **Pre/Post Questionnaires:** Attach or enable any required demographic, consent, or psychometric surveys to be administered immediately before or after the main VLAB task.
- **Experiment KEY:** Generate or retrieve the unique access code for this specific study. *Note: This is the vital authentication key you must distribute to your participants, allowing them to load your specific experiment upon launching the application.*



To add your questions:

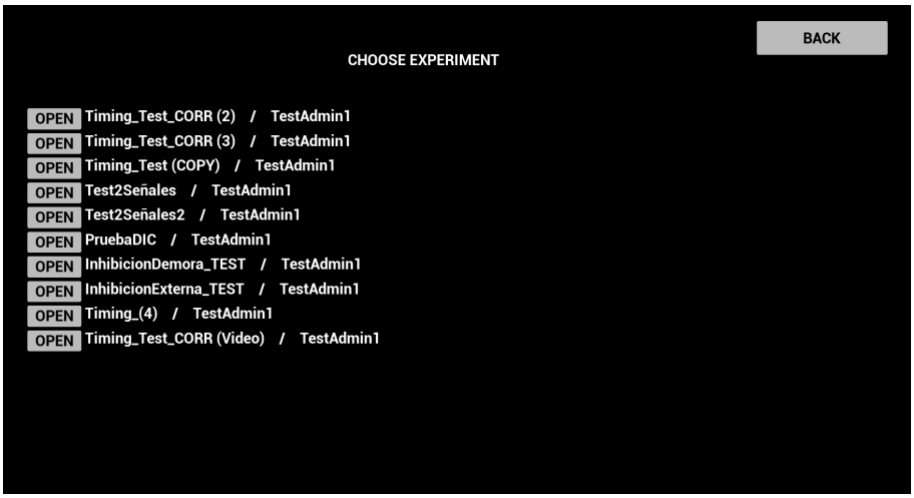
1. Click the **Add Question** text field (or corresponding button).
2. Type your Yes/No question exactly as you want it to appear to the participant.
3. Click **Add Question** to lock the actual question into the questionnaire sequence. Repeat this process for all remaining survey items until you **Finished**.



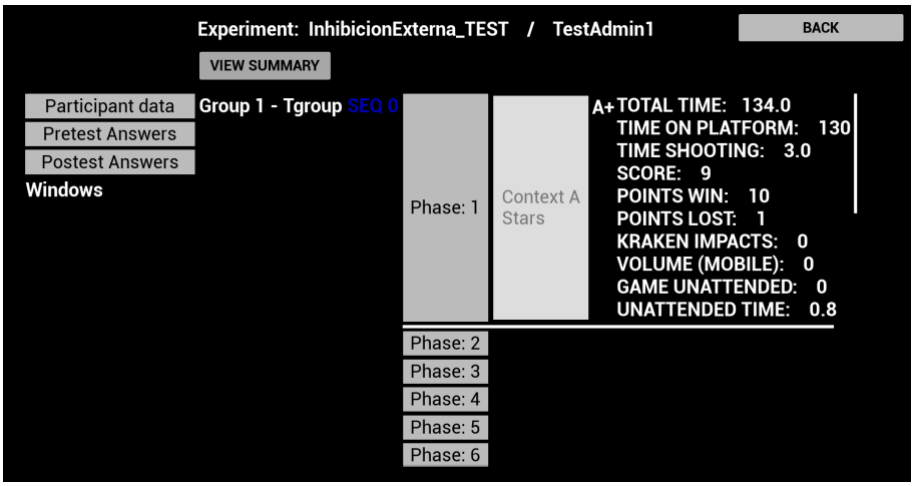
2. See My Results

The Experiment Directory

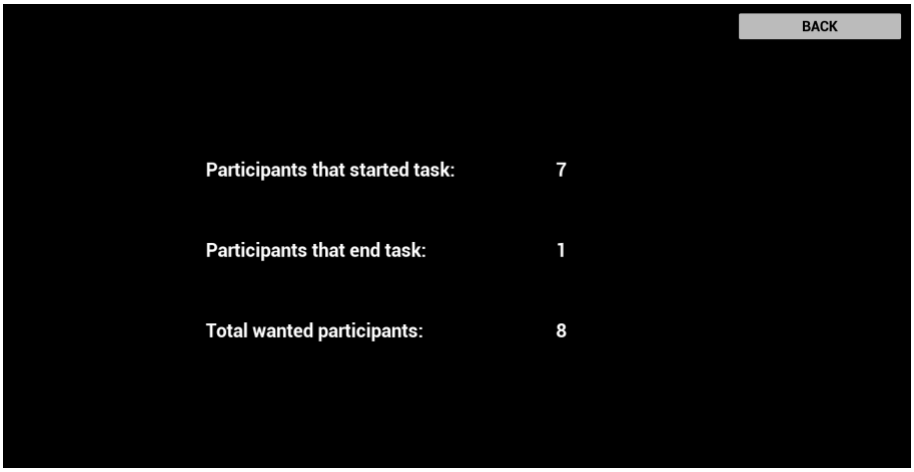
Upon navigating to this menu, the system will actively populate a list of all experimental configurations associated with your unique Experimenter Key. From this dashboard, you can view, select, and manage any of your previously saved designs.



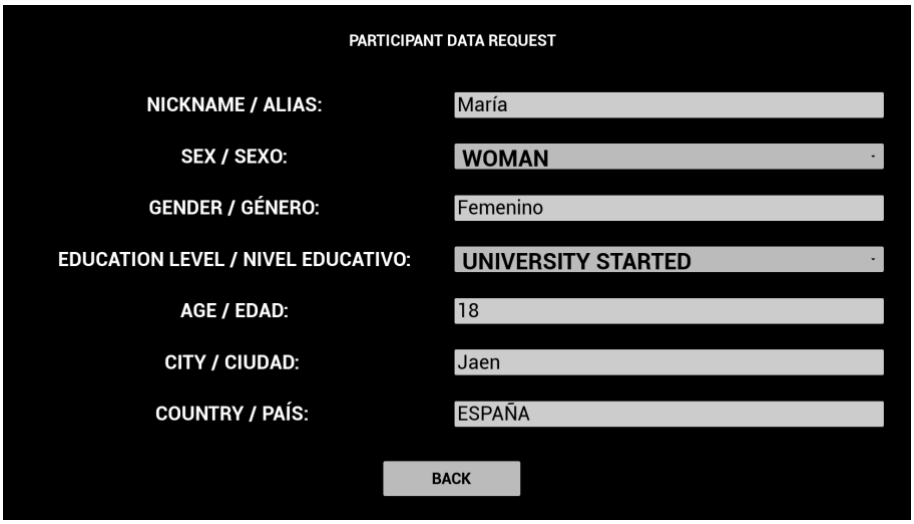
Viewing Results: Selecting a specific experiment from the list will open its dedicated data dashboard. From here, you can review core participant demographic profiles alongside their associated trial data, including both fully completed sessions and partial, in-progress results.



Viewing Aggregate Data: By clicking the **VIEW SUMMARY** button, you can review the calculated performance averages across your participants.



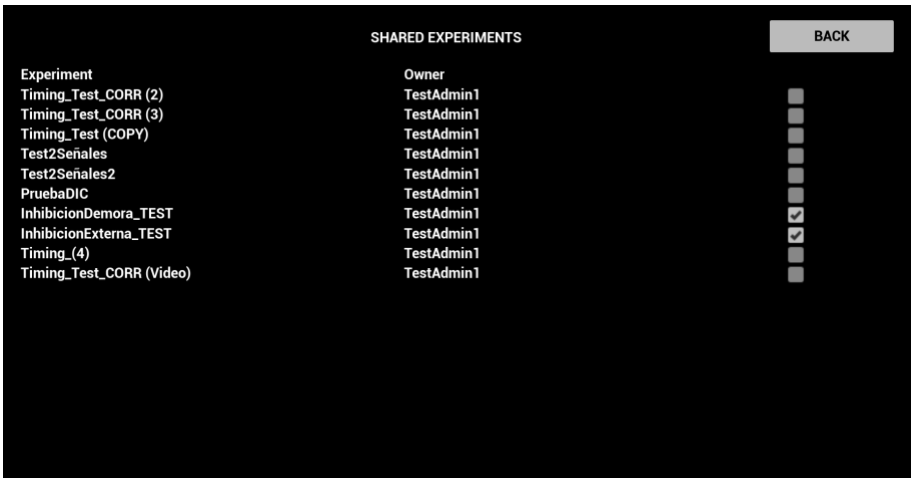
Participant Data: > Select the **Participant Data** button to view individual demographic profiles and specific trial results.



Reviewing Questionnaire Data: To view a specific participant's responses to the assigned surveys, click the respective **Pretest** or **Posttest** buttons located next to their profile.

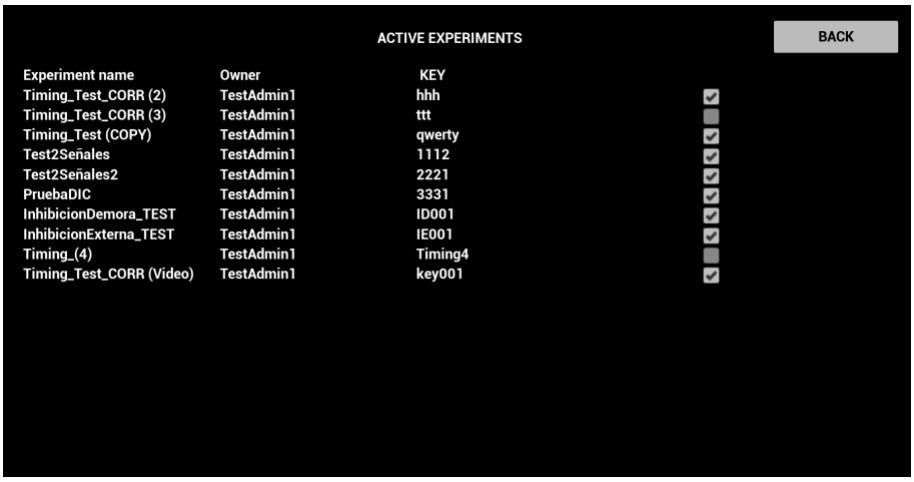
3. Share Experiments

Sharing Toggle: Each experiment in your list features a dedicated sharing checkbox. Checking this box publishes the task to the wider VLAB community, while unchecking it instantly revokes access and returns the experiment to private status.



4. Disable Experiments

Enabling and Disabling Experiments: You can control active access to your tasks using the **Enable/Disable** checkbox. Unchecking this box deactivates the experiment, preventing any new participants from joining or launching the session.



Participant's Guide

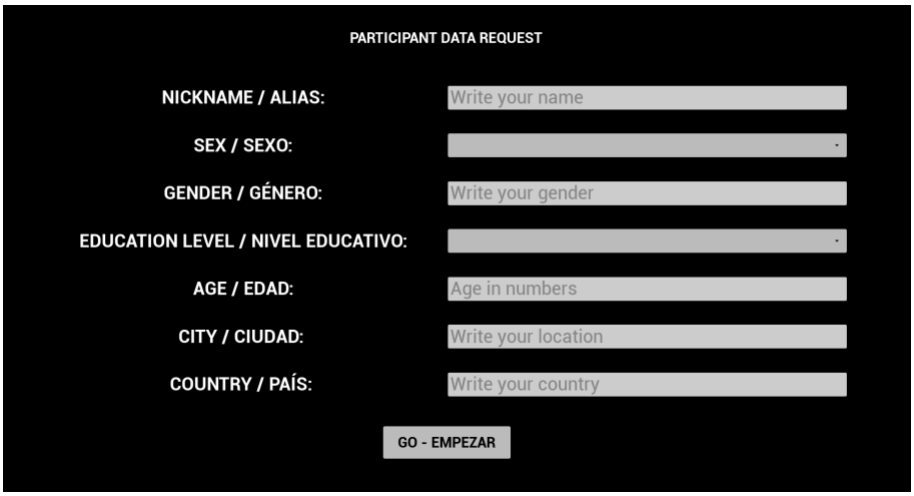
Accessing an Experiment

1. Enter the Experiment Key Upon launching the VLAB application, you will be greeted by the initial login screen. In the designated text field, type the unique **Experiment KEY** that was provided to you by your researcher or instructor. Once entered, click the confirmation button to load your specific task.

2. Complete Your Demographic Profile After successfully validating the Experiment Key, you will be directed to a demographic intake screen. Please fill in the requested fields using the provided text boxes and dropdown menus. You will be asked to provide:

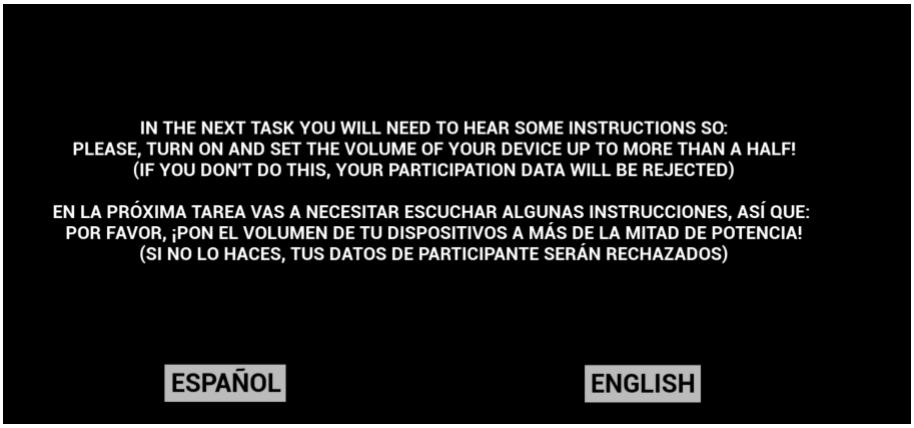
- **Nickname:** Please enter an anonymous alias or the specific Participant ID assigned to you by the researcher. *Do not use your real name.*
- **Sex & Gender:** Your biological sex and current gender identity.
- **Age:** Your current age in years.
- **Education:** Your highest level of completed schooling.
- **Location & Country:** Your current city/region and country of residence.

Note: All demographic data is collected strictly for research purposes and is stored securely and anonymously.



3. Complete the Pre-Task Questionnaire (If Applicable) If the researcher has assigned a preliminary survey to your session, it will appear on screen now. Please read each prompt carefully and answer the series of Yes/No questions using the provided on-screen toggles. If no questionnaire was assigned, the software will automatically skip this step.

4. Configure Audio and Review Instructions The upcoming gamified task may utilize important auditory cues and warning signals. Please ensure your device's volume is turned up or your headphones are securely connected. Once your audio is set, select your preferred language by clicking either the **English** or **Español** button. This will play the audio instructions and display the text briefing for your specific task. Please review these rules carefully before proceeding.



5. Press Go, and start task

Task Instructions

Welcome to the VLAB virtual laboratory. The task you are about to perform is a spaceship game where the goal is to earn as many points as you can. You will see a dotted marker in the top center of the screen. By putting your finger on the screen without lifting it (or using your mouse), you will drive the ship while shooting at enemies. Each enemy killed will earn you points, but occasionally, an indestructible enemy will appear and deduct points from you massively. To avoid this and conserve your points, try to anticipate its appearance by staying safe until it leaves. To do this, you have a green platform on each side that acts as a safe place. No enemy can hurt you here, but beware—you won't be able to earn points while on the platform either! Good luck and enjoy the game.